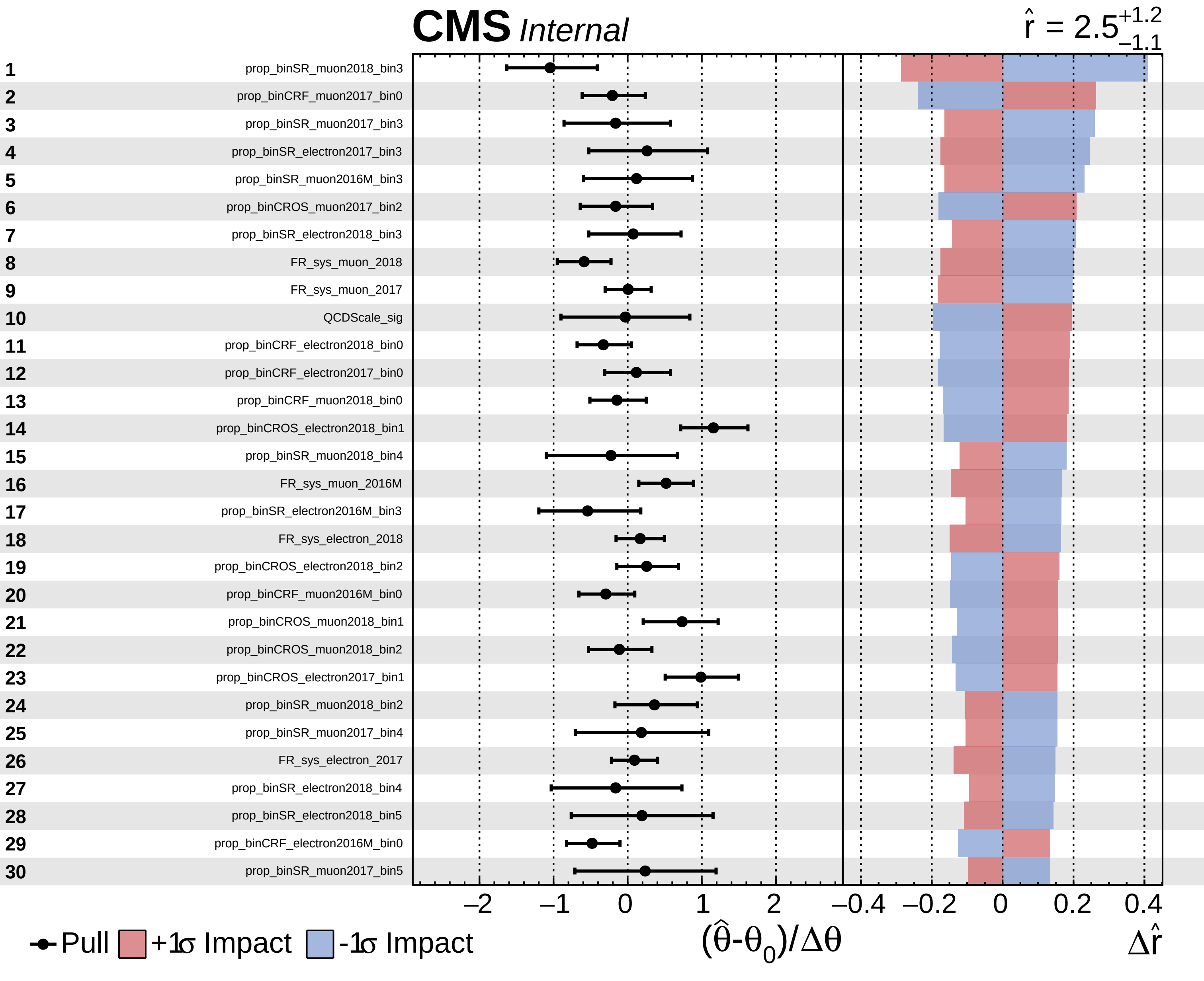
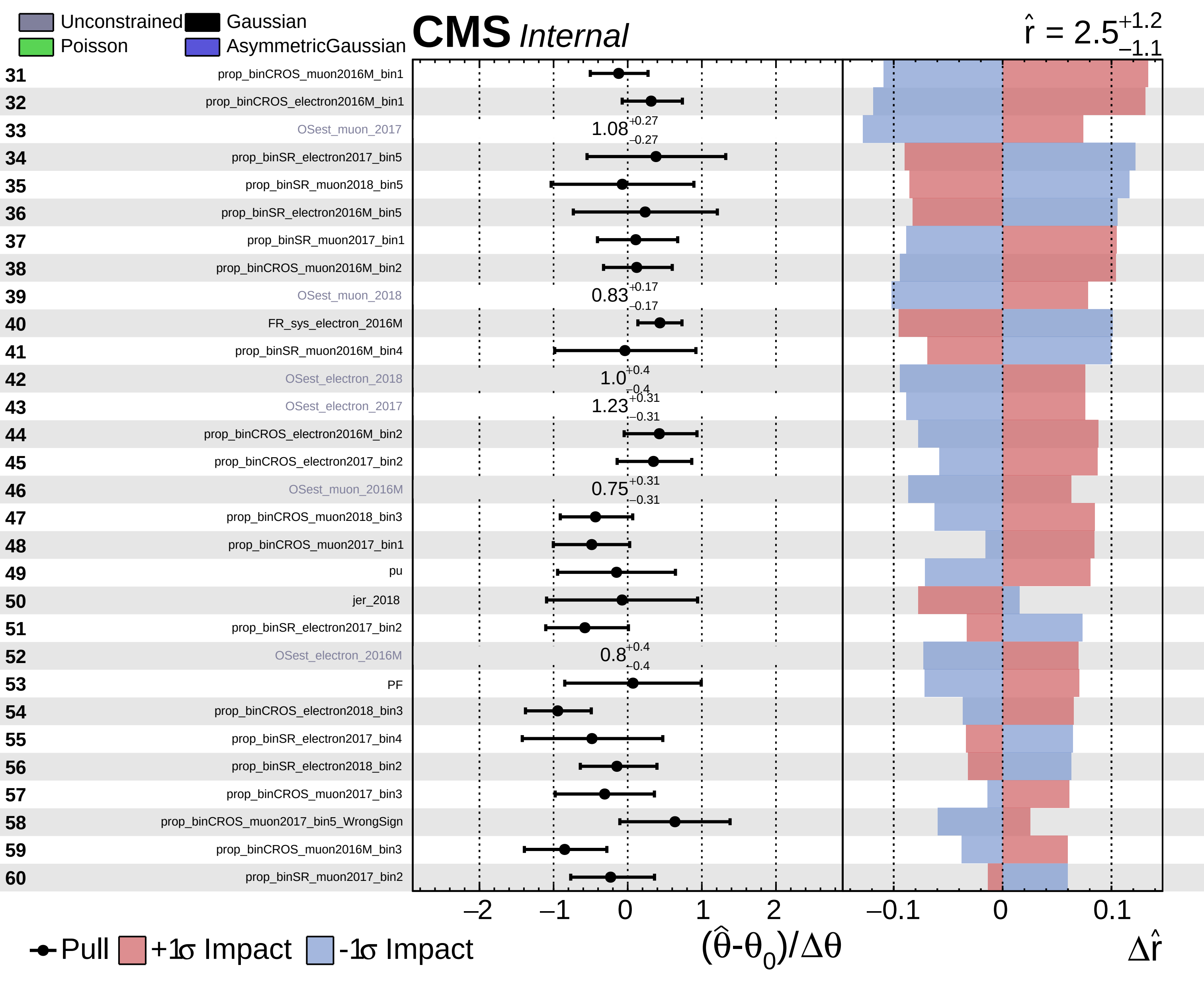
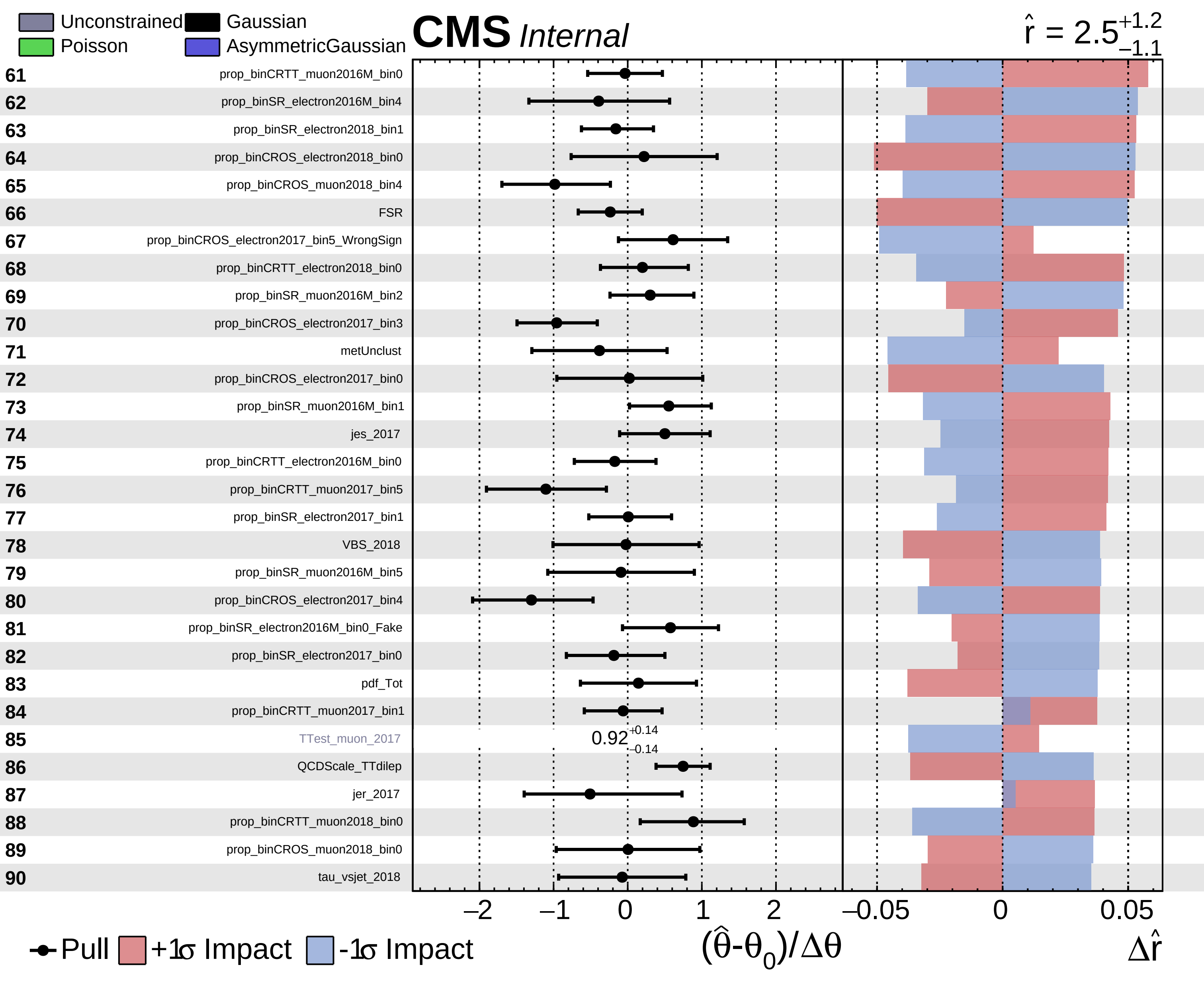
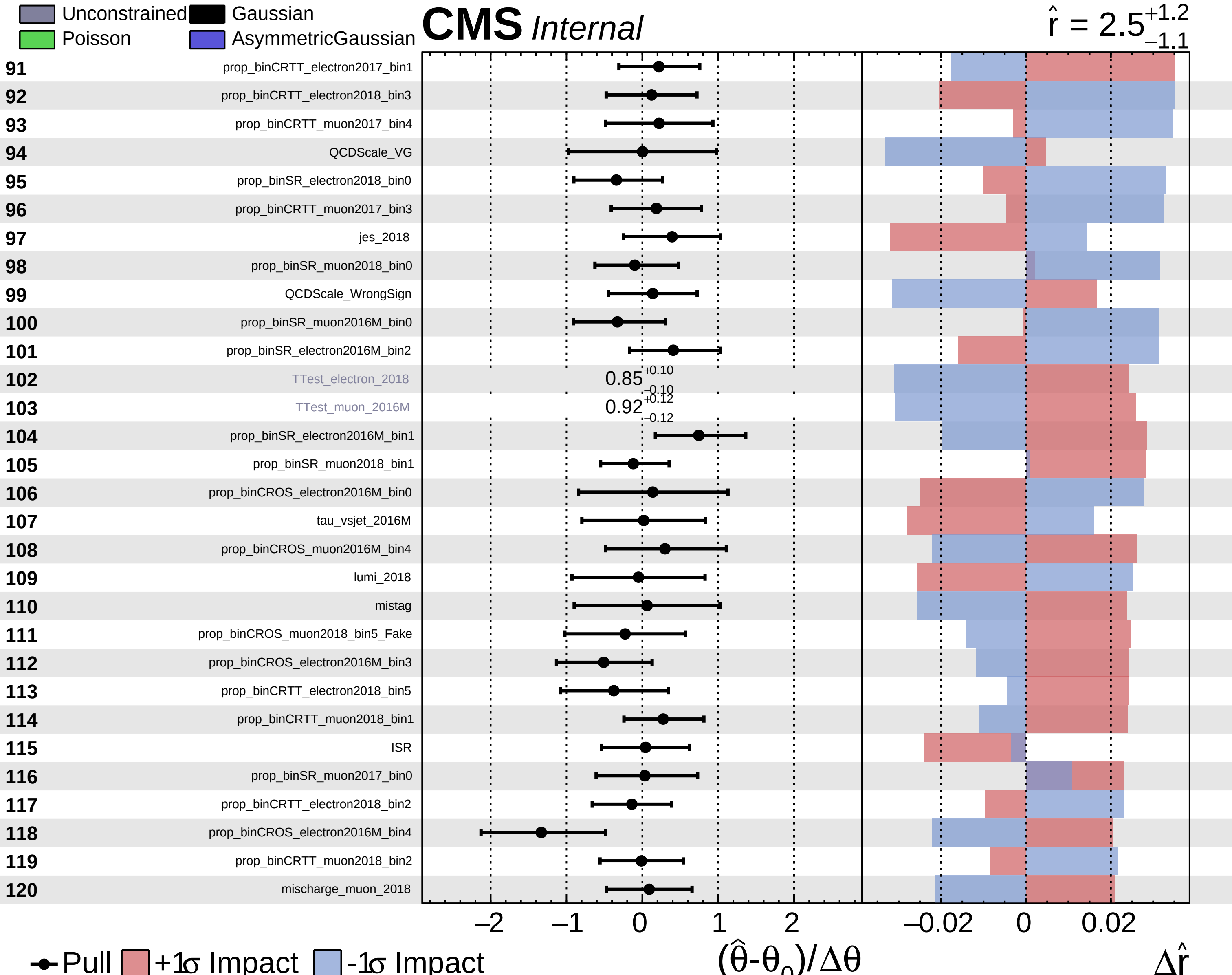
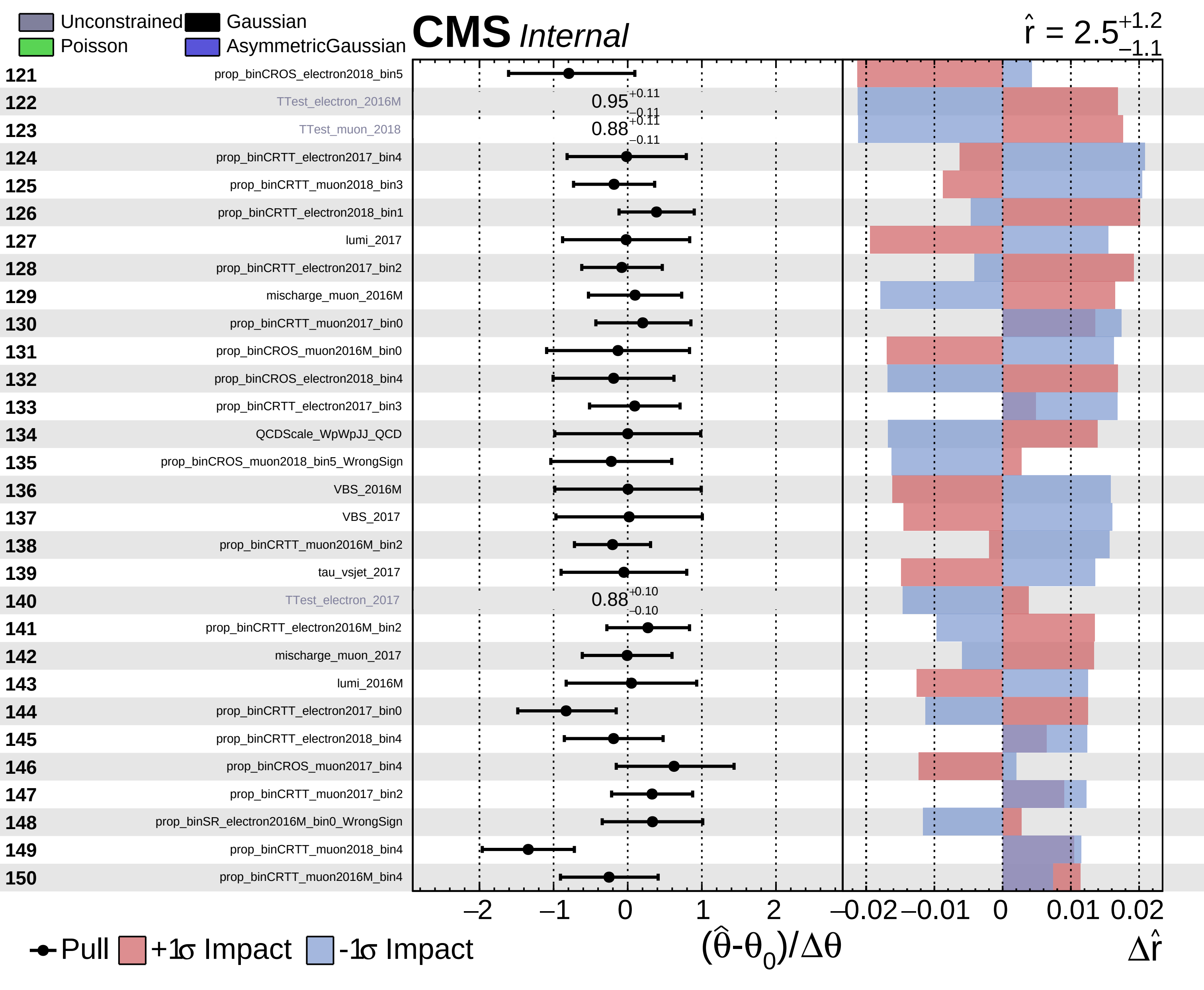

CMS Internal
$\hat{r} = 2.5^{+1.2}_{-1.1}$
1 prop_binSR_muon2018_bin3
2 prop_binCRF_muon2017_bin0
3 prop_binSR_muon2017_bin3
4 prop_binSR_electron2017_bin3
5 prop_binSR_muon2016M_bin3
6 prop_binCROS_muon2017_bin2
7 prop_binSR_electron2018_bin3
8 FR_sys_muon_2018
9 FR_sys_muon_2017
10 QCDScale_sig
11 prop_binCRF_electron2018_bin0
12 prop_binCRF_electron2017_bin0
13 prop_binCRF_muon2018_bin0
14 prop_binCROS_electron2018_bin1
15 prop_binSR_muon2018_bin4
16 FR_sys_muon_2016M
17 prop_binSR_electron2016M_bin3
18 FR_sys_electron_2018
19 prop_binCROS_electron2018_bin2
20 prop_binCRF_muon2016M_bin0
21 prop_binCROS_muon2018_bin1
22 prop_binCROS_muon2018_bin2
23 prop_binCROS_electron2017_bin1
24 prop_binSR_muon2018_bin2
25 prop_binSR_muon2017_bin4
26 FR_sys_electron_2017
27 prop_binSR_electron2018_bin4
28 prop_binSR_electron2018_bin5
29 prop_binCRF_electron2016M_bin0
30 prop_binSR_muon2017_bin5
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



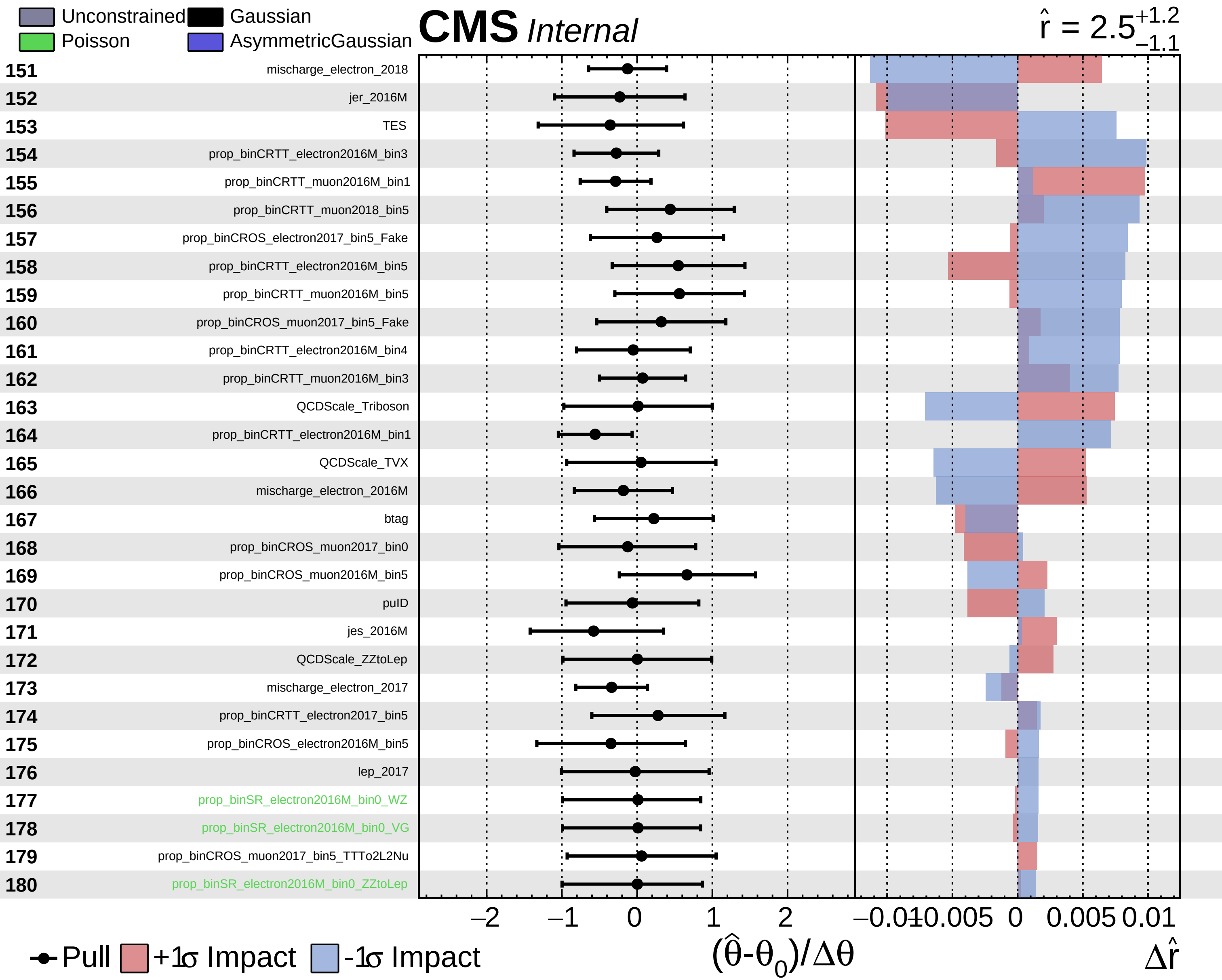
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 2.5 +1.2 −1.1
31 prop_binCROS_muon2016M_bin1
32 prop_binCROS_electron2016M_bin1
33 OSest_muon_2017
1.08 +0.27 −0.27
34 prop_binSR_electron2017_bin5
35 prop_binSR_muon2018_bin5
36 prop_binSR_electron2016M_bin5
37 prop_binSR_muon2017_bin1
38 prop_binCROS_muon2016M_bin2
39 OSest_muon_2018
0.83 +0.17 −0.17
40 FR_sys_electron_2016M
41 prop_binSR_muon2016M_bin4
42 OSest_electron_2018
1.0 +0.4 −0.4
43 OSest_electron_2017
1.23 +0.31 −0.31
44 prop_binCROS_electron2016M_bin2
45 prop_binCROS_electron2017_bin2
46 OSest_muon_2016M
0.75 +0.31 −0.31
47 prop_binCROS_muon2018_bin3
48 prop_binCROS_muon2017_bin1
49 pu
50 jer_2018
51 prop_binSR_electron2017_bin2
52 OSest_electron_2016M
0.8 +0.4 −0.4
53 PF
54 prop_binCROS_electron2018_bin3
55 prop_binSR_electron2017_bin4
56 prop_binSR_electron2018_bin2
57 prop_binCROS_muon2017_bin3
58 prop_binCROS_muon2017_bin5_WrongSign
59 prop_binCROS_muon2016M_bin3
60 prop_binSR_muon2017_bin2
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

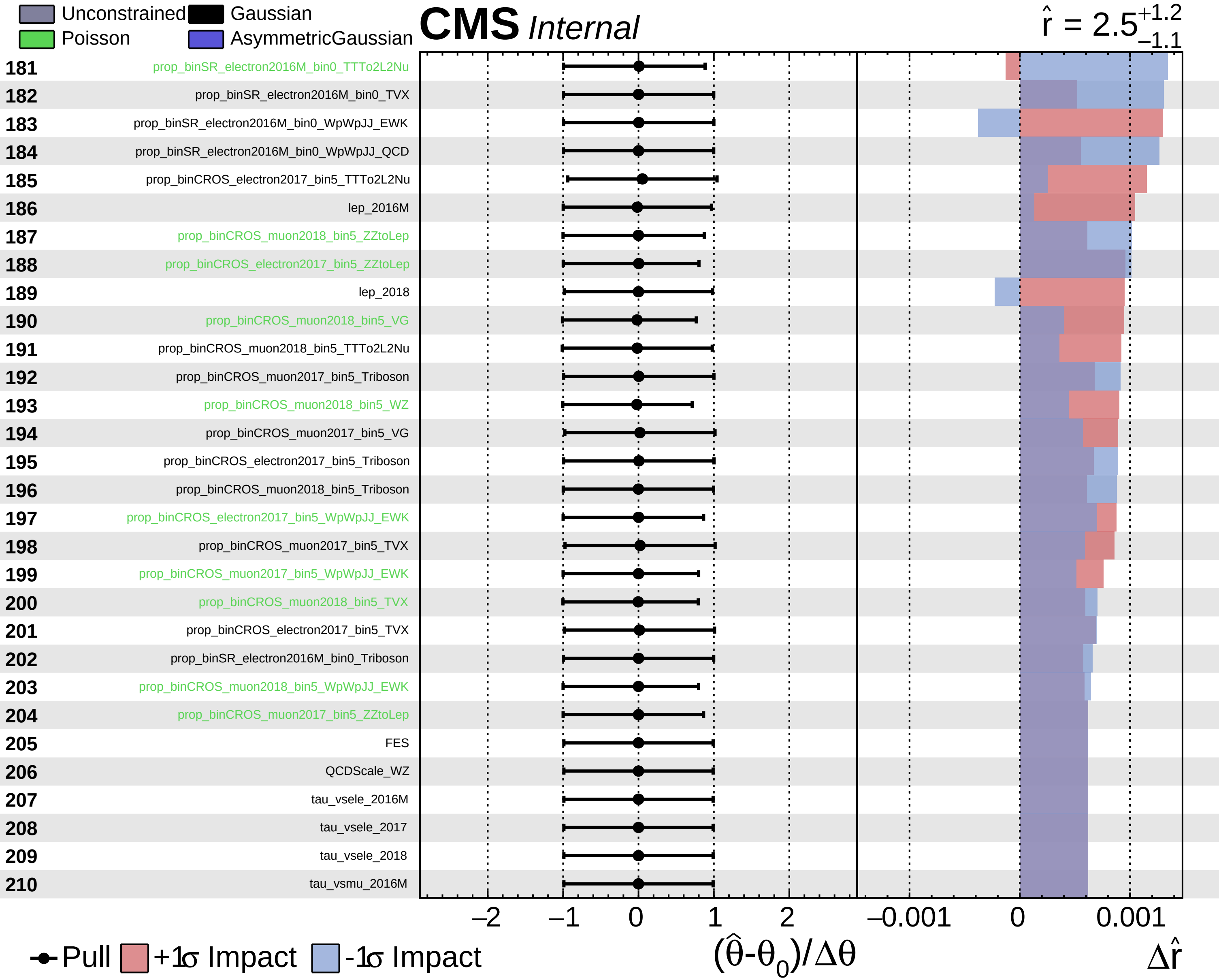

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.5^{+1.2}_{-1.1}$
61 prop_binCRTT_muon2016M_bin0
62 prop_binSR_electron2016M_bin4
63 prop_binSR_electron2018_bin1
64 prop_binCROS_electron2018_bin0
65 prop_binCROS_muon2018_bin4
66 FSR
67 prop_binCROS_electron2017_bin5_WrongSign
68 prop_binCRTT_electron2018_bin0
69 prop_binSR_muon2016M_bin2
70 prop_binCROS_electron2017_bin3
71 metUnclust
72 prop_binCROS_electron2017_bin0
73 prop_binSR_muon2016M_bin1
74 jes_2017
75 prop_binCRTT_electron2016M_bin0
76 prop_binCRTT_muon2017_bin5
77 prop_binSR_electron2017_bin1
78 VBS_2018
79 prop_binSR_muon2016M_bin5
80 prop_binCROS_electron2017_bin4
81 prop_binSR_electron2016M_bin0_Fake
82 prop_binSR_electron2017_bin0
83 pdf_Tot
84 prop_binCRTT_muon2017_bin1
85 TTest_muon_2017
$0.92^{+0.14}_{-0.14}$
86 QCDScale_TTdilep
87 jer_2017
88 prop_binCRTT_muon2018_bin0
89 prop_binCROS_muon2018_bin0
90 tau_vsjet_2018
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

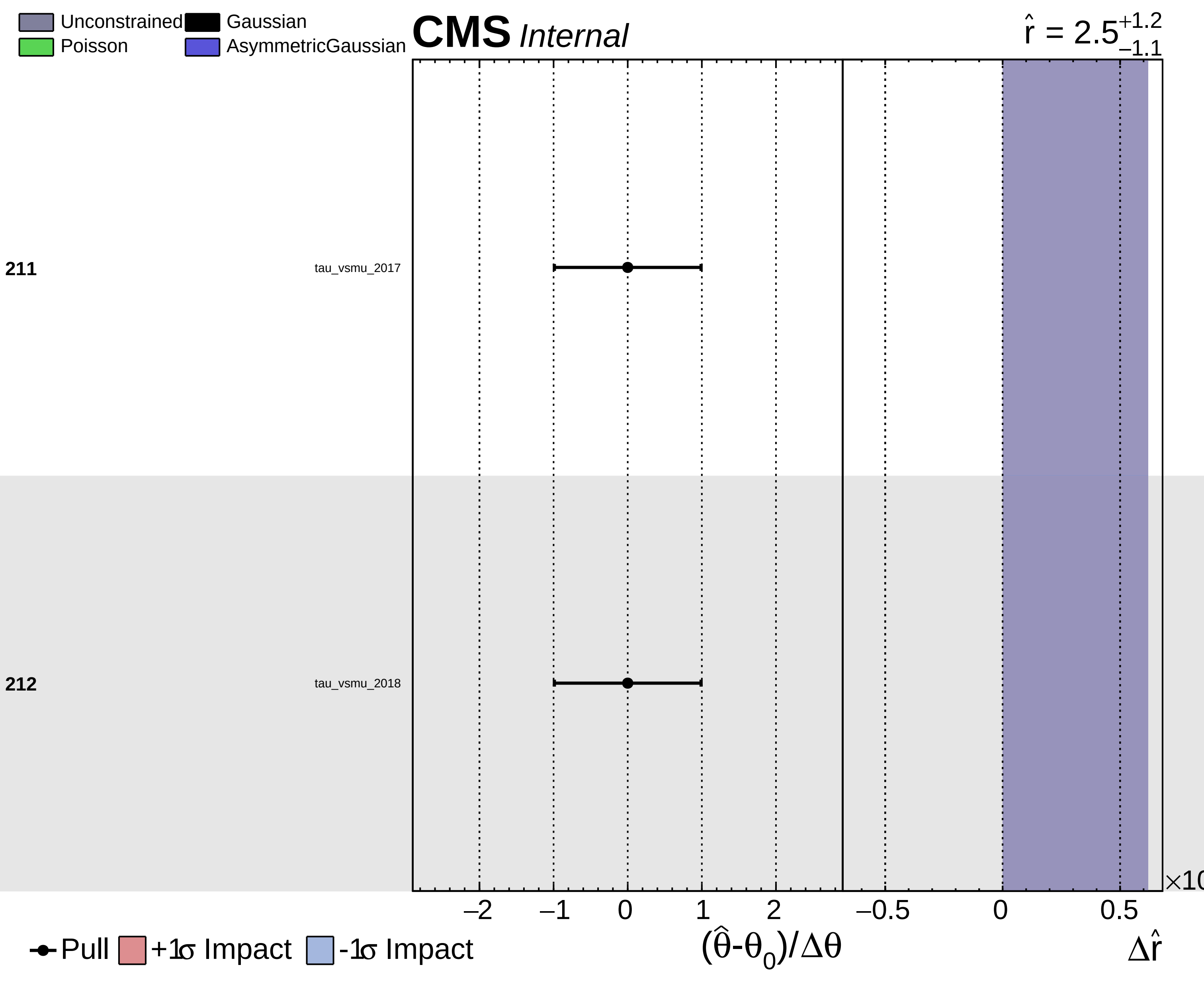

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.5^{+1.2}_{-1.1}$
91 prop_binCRTT_electron2017_bin1
92 prop_binCRTT_electron2018_bin3
93 prop_binCRTT_muon2017_bin4
94 QCDScale_VG
95 prop_binSR_electron2018_bin0
96 prop_binCRTT_muon2017_bin3
97 jes_2018
98 prop_binSR_muon2018_bin0
99 QCDScale_WrongSign
100 prop_binSR_muon2016M_bin0
101 prop_binSR_electron2016M_bin2
102 TTest_electron_2018 $0.85^{+0.10}_{-0.10}$
103 TTest_muon_2016M $0.92^{+0.12}_{-0.12}$
104 prop_binSR_electron2016M_bin1
105 prop_binSR_muon2018_bin1
106 prop_binCROS_electron2016M_bin0
107 tau_vsjet_2016M
108 prop_binCROS_muon2016M_bin4
109 lumi_2018
110 mistag
111 prop_binCROS_muon2018_bin5_Fake
112 prop_binCROS_electron2016M_bin3
113 prop_binCRTT_electron2018_bin5
114 prop_binCRTT_muon2018_bin1
115 ISR
116 prop_binSR_muon2017_bin0
117 prop_binCRTT_electron2018_bin2
118 prop_binCROS_electron2016M_bin4
119 prop_binCRTT_muon2018_bin2
120 mischarge_muon_2018
−2 −1 0 1 2
−0.02 0 0.02
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.5^{+1.2}_{-1.1}$
121 prop_binCROS_electron2018_bin5
122 TTest_electron_2016M $0.95^{+0.11}_{-0.11}$
123 TTest_muon_2018 $0.88^{+0.11}_{-0.11}$
124 prop_binCRTT_electron2017_bin4
125 prop_binCRTT_muon2018_bin3
126 prop_binCRTT_electron2018_bin1
127 lumi_2017
128 prop_binCRTT_electron2017_bin2
129 mischarge_muon_2016M
130 prop_binCRTT_muon2017_bin0
131 prop_binCROS_muon2016M_bin0
132 prop_binCROS_electron2018_bin4
133 prop_binCRTT_electron2017_bin3
134 QCDScale_WpWpJJ_QCD
135 prop_binCROS_muon2018_bin5_WrongSign
136 VBS_2016M
137 VBS_2017
138 prop_binCRTT_muon2016M_bin2
139 tau_vsjet_2017
140 TTest_electron_2017 $0.88^{+0.10}_{-0.10}$
141 prop_binCRTT_electron2016M_bin2
142 mischarge_muon_2017
143 lumi_2016M
144 prop_binCRTT_electron2017_bin0
145 prop_binCRTT_electron2018_bin4
146 prop_binCROS_muon2017_bin4
147 prop_binCRTT_muon2017_bin2
148 prop_binSR_electron2016M_bin0_WrongSign
149 prop_binCRTT_muon2018_bin4
150 prop_binCRTT_muon2016M_bin4
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.5^{+1.2}_{-1.1}$
151 mischarge_electron_2018
152 jer_2016M
153 TES
154 prop_binCRTT_electron2016M_bin3
155 prop_binCRTT_muon2016M_bin1
156 prop_binCRTT_muon2018_bin5
157 prop_binCROS_electron2017_bin5_Fake
158 prop_binCRTT_electron2016M_bin5
159 prop_binCRTT_muon2016M_bin5
160 prop_binCROS_muon2017_bin5_Fake
161 prop_binCRTT_electron2016M_bin4
162 prop_binCRTT_muon2016M_bin3
163 QCDScale_Triboson
164 prop_binCRTT_electron2016M_bin1
165 QCDScale_TVX
166 mischarge_electron_2016M
167 btag
168 prop_binCROS_muon2017_bin0
169 prop_binCROS_muon2016M_bin5
170 puID
171 jes_2016M
172 QCDScale_ZZtoLep
173 mischarge_electron_2017
174 prop_binCRTT_electron2017_bin5
175 prop_binCROS_electron2016M_bin5
176 lep_2017
177 prop_binSR_electron2016M_bin0_WZ
178 prop_binSR_electron2016M_bin0_VG
179 prop_binCROS_muon2017_bin5_TTTo2L2Nu
180 prop_binSR_electron2016M_bin0_ZZtoLep
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 2.5 +1.2 −1.1
181 prop_binSR_electron2016M_bin0_TTTo2L2Nu
182 prop_binSR_electron2016M_bin0_TVX
183 prop_binSR_electron2016M_bin0_WpWpJJ_EWK
184 prop_binSR_electron2016M_bin0_WpWpJJ_QCD
185 prop_binCROS_electron2017_bin5_TTTo2L2Nu
186 lep_2016M
187 prop_binCROS_muon2018_bin5_ZZtoLep
188 prop_binCROS_electron2017_bin5_ZZtoLep
189 lep_2018
190 prop_binCROS_muon2018_bin5_VG
191 prop_binCROS_muon2018_bin5_TTTo2L2Nu
192 prop_binCROS_muon2017_bin5_Triboson
193 prop_binCROS_muon2018_bin5_WZ
194 prop_binCROS_muon2017_bin5_VG
195 prop_binCROS_electron2017_bin5_Triboson
196 prop_binCROS_muon2018_bin5_Triboson
197 prop_binCROS_electron2017_bin5_WpWpJJ_EWK
198 prop_binCROS_muon2017_bin5_TVX
199 prop_binCROS_muon2017_bin5_WpWpJJ_EWK
200 prop_binCROS_muon2018_bin5_TVX
201 prop_binCROS_electron2017_bin5_TVX
202 prop_binSR_electron2016M_bin0_Triboson
203 prop_binCROS_muon2018_bin5_WpWpJJ_EWK
204 prop_binCROS_muon2017_bin5_ZZtoLep
205 FES
206 QCDScale_WZ
207 tau_vsele_2016M
208 tau_vsele_2017
209 tau_vsele_2018
210 tau_vsmu_2016M
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.5^{+1.2}_{-1.1}$
211
tau_vsmu_2017
212
tau_vsmu_2018
–2
–1
0
1
2
–0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$